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CLASS
INDEX NO.

CANDIDATE NAME: _____

07 _____

TAMPINES JUNIOR COLLEGE PRELIMINARY EXAMINATION 2008

BIOLOGY Paper 1 Multiple Choice

8875/01**TUESDAY 26 AUGUST 2008****12.30 – 1.30 PM (1 hour)**

Additional materials: Multiple Choice Answer Sheet

INSTRUCTIONS TO CANDIDATES

1. For 'Index Number', shade the **last 5 numbers** of your **NRIC** and shade letter 'A' for local students and letter 'F' for foreign students on the **Optical Answer Sheet** in the spaces designated.
2. Write in soft pencil.
3. Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

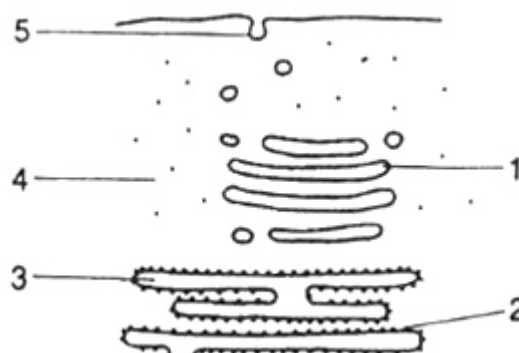
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Calculations may be used.

This question paper consists of **14** printed pages.

1. Radioactive amino acids are supplied to a cell that uses them to make insulin.



Which route will the radioactive amino acids take?

	First				Last
A	4	2	3	1	5
B	4	3	2	1	5
C	5	1	3	2	4
D	5	3	2	4	1

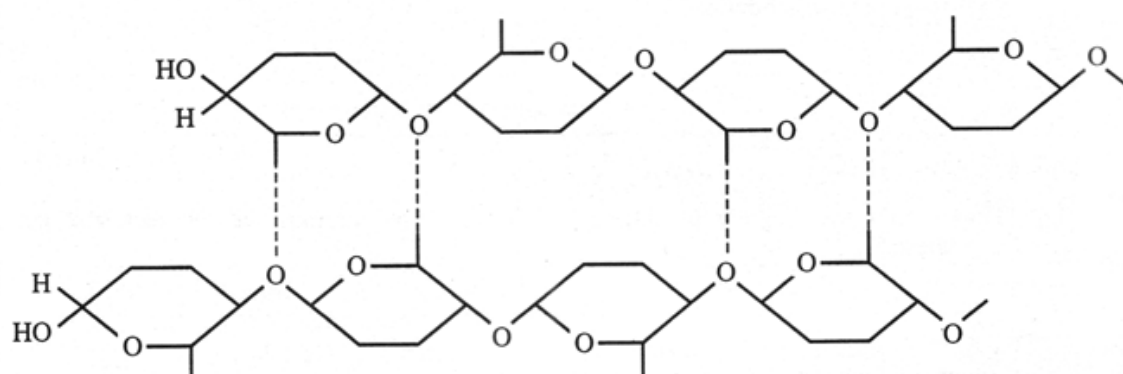
2. What could be a possible explanation for the ability of lysosomes to withstand self-digestion?
 - A lysosomes contain inactive hydrolytic enzymes
 - B lysosome membrane has numerous modified proteins with carbohydrate side-chains
 - C lysosomes do contain lipases which are the enzymes capable of digesting the lipid membrane
 - D hydrolases in the lysosomes are inhibited by the acidic internal environment of the lysosomes
3. A certain cell surface membrane is made up entirely of phospholipids and is 5nm thick. A volume of 1 mm^3 of this tray of phospholipids membrane was homogenized and dropped onto the surface of water in a large tray. The phospholipids spread out to form a continuous thin film.

What is the expected area of this film?

- A $200\,000 \text{ mm}^2$ because the molecules form a single layer
- B $200\,000 \text{ mm}^2$ because the molecules form a double layer
- C $400\,000 \text{ mm}^2$ because the molecules form a single layer
- D $400\,000 \text{ mm}^2$ because the molecules form a double layer



4. The figure below shows part of the molecular structure of a polysaccharide.



Key: - - - hydrogen bonds

Which of the following statement(s) is/are **true**?

- I The structure contains β -glucose.
 - II The polysaccharide, which is an important food source for animals, can be digested by humans.
 - III The polysaccharide may associate in groups to form microfibrils, which are arranged in larger bundles to form macrofibrils.
- A I and II only
 - B II and III only
 - C I and III only
 - D All of the above

5. The table shows some differences between glucose and triglyceride. Which difference explains the fact that glucose is soluble in water, while triglyceride is insoluble in water?

<i>Glucose</i>	<i>Triglyceride</i>
A Contains polar groups	Does not contain polar groups
B Does not contain double bonds	Contains double bonds
C Higher ratio of carbon:hydrogen	Lower ratio of carbon:hydrogen
D Occurs in a ring form	Does not occur in a ring form



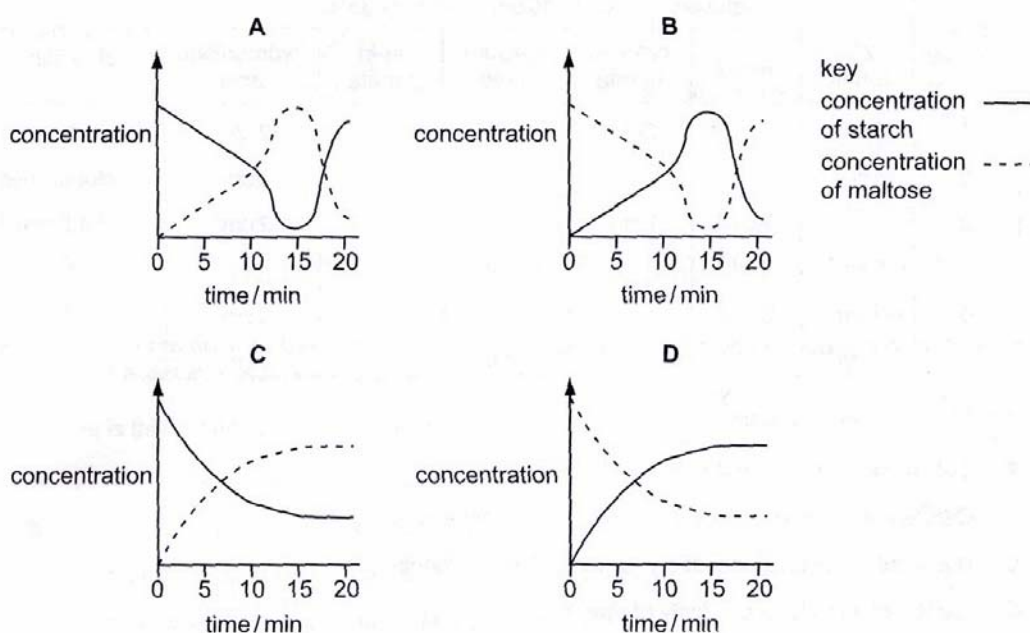
6. The diagram shows the 3D conformation of a protein.



What would happen when this protein denatures?

- A** It will be hydrolyzed into its component amino acids.
 - B** All the α -helices and β -pleated conformation will be lost.
 - C** It loses its secondary and tertiary structures.
 - D** Its hydrogen bonds, ionic bonds, hydrophobic interactions, disulphide linkages as well as peptide bonds are disrupted.
7. In an experiment using starch and amylase, the concentrations of starch and maltose present in the reacting mixture are measured every minute for 20 minutes. 1% hydrochloric acid is added after 10 minutes and the mixture is heated to 60°C at 14 minutes.

Which graph represents the results of this experiment?



8. The concentration of carbon dioxide in a sample of air was found to be 280 ppm (parts per million). A controlled experiment was designed to measure the concentration of carbon dioxide in the air after it had flowed over the leaves of a green plant. Measurements were taken at a range of light intensities.

The following results were obtained:

Light intensity (% of full sunlight)	Concentration of carbon dioxide in air after flowing over leaves (ppm)
75	253
50	252
25	254
10	280

Which one of the following statements is **not** consistent with these results?

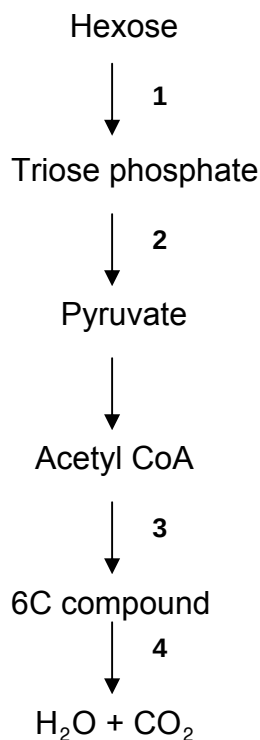
A	At the lower light intensities tested, the rate of photosynthesis is limited by light intensity.
B	At the higher light intensities tested, the rate of photosynthesis is affected by factors other than light.
C	In the dark, the concentration of carbon dioxide in the air after it had flowed over leaves would be at least 280 ppm.
D	At a light intensity of 10% of full sunlight, photosynthesis does not occur.

9. Some photosynthetic organisms containing chloroplasts that lack PS II (photosystem II) are able to survive. The best way to detect the lack of PS II in these organisms would be to;

- A** determine if they have thylakoids in the chloroplasts.
- B** test for liberation of oxygen in the light.
- C** test for carbon fixation in the dark.
- D** determine the production of starch.



10. The diagram summarizes the pathway of glucose breakdown.



Which two steps results in a net increase of ATP?

- A** 1 and 3.
- B** 1 and 4.
- C** 2 and 3.
- D** 2 and 4.

11. The events shown below occur during different phases of mitosis.

- 1. Spiralisation of DNA
- 2. Reappearance of nuclear membrane
- 3. Centromeres split
- 4. Centromeres attach to spindle fibres
- 5. DNA replicates

Which of the following correctly identifies each of the phases described?

	Interphase	Prophase	Metaphase	Anaphase	Telophase
A	1	2	3	4	5
B	5	1	4	3	2
C	3	2	4	1	5
D	4	5	1	2	3



12. Which of the following statements is **true** regarding point mutations?

- 1 They could result in a frameshift mutation.
 - 2 They always produce a change in the amino acid sequence of a protein.
 - 3 They could result in genes being duplicated.
- A** 1 only
B 1 and 2
C 2 and 3
D 1, 2 and 3

13. A new form of life is discovered. It has a genetic code much like that of organisms on Earth except that there are **five** different DNA bases instead of four and the base sequences are translated as **doublets** instead of triplets. How many different amino acids could be accommodated by this genetic code?

- A** 25
B 32
C 10
D 64

14. The table below shows some of the events which take place in protein synthesis.

- 1 tRNA molecules bring specific amino acids to the ribosome.
- 2 Ribonucleotides join with exposed DNA bases and form a molecule of mRNA.
- 3 Release factor binds to mRNA.
- 4 Peptide bonds formed between amino acids.
- 5 mRNA molecule leaves the nucleus.
- 6 mRNA undergoes capping and addition of poly(A) tail.

Which of the following shows the correct sequence of events?

- A** 2 5 6 1 4 3
B 2 6 5 1 4 3
C 2 6 5 1 3 4
D 2 5 6 1 3 4



15. A biochemist isolates and purifies various molecules needed for DNA replication. When she adds some DNA, replication occurs but the DNA molecules formed are defective. Each consists of a normal DNA strand paired with numerous segments of DNA a few hundred nucleotides long. What has she probably left out of the mixture?

- A DNA polymerase
- B nucleotides
- C primers
- D ligase

16. Identify the **CORRECT** statement.

- A A promoter sequence region (for transcription) in a bacterial cell is normally transcribed into the primary transcript.
- B In mammalian cells, promoter sequences are normally located at the 3' end of a gene.
- C In mammalian cells, transcribed sequences never participate in promoter function.
- D In bacterial cells, the promoter region is normally located at the 5' end of a gene.

17. Enhancers of gene expression are;

- A proteins that regulate spatial and temporal control of genes.
- B regulatory DNA sequences.
- C used for expression of all genes.
- D never found in eukaryotes.

18. Which of the following is **not** true of telomerase?

- A Telomerase contains a ribozyme.
- B Telomerase activity decreases as the cell ages.
- C Telomerase adds a repeating sequence to the 5' end of DNA.
- D Telomerase adds a repeating sequence to the 3' end of DNA.



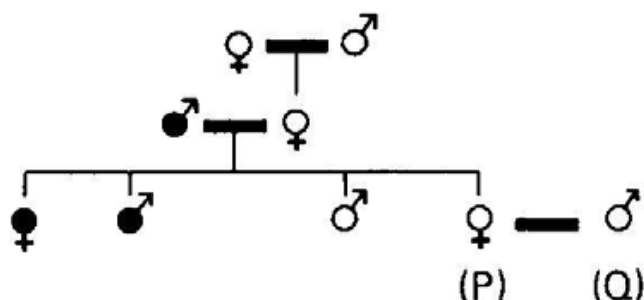
19. The table shows a comparison between the genomes of a prokaryote and eukaryote.

	Prokaryote	Eukaryote
Genome length (base pairs)	4 640 000	12 068 000
Number of proteins	4300	6200
Proteins with roles in metabolism	650	650
Proteins with roles in energy release	240	175
Proteins with roles in protein synthesis	410	750

Which feature of eukaryotes accounts for the differences in the number of proteins with roles in protein synthesis?

- A The DNA of a eukaryote has histones.
- B The DNA of a eukaryote has introns.
- C The DNA of a eukaryote has more base pairs.
- D The DNA of a eukaryote is separated from the ribosomes by membranes.

20. In a species where the female is homogametic, a sex-linked allelic pair of genes controls pigmentation. The following results were obtained during the course of a breeding experiment.



- ♂ ♀ = individuals with normal pigmentation
 ♂ ♀ = individuals with abnormal pigmentation
 — = a 'mating line'

Which of the following ratios of offspring will be produced when P and Q are bred together?

- A 2 females (both normal) : 2 males (both abnormal)
- B 2 females (both carriers) : 2 males (both normal)
- C 2 females (1 carrier, 1 abnormal) : 2 males (1 normal, 1 abnormal)
- D 2 females (1 carrier, 1 normal) : 2 males (1 normal, 1 abnormal)



21. The condition in which a zygote contains three copies of a particular chromosome as a result of nondisjunction is called _____.

- A trisomy
- B monosomy
- C polyploidy
- D monoploidy

22. The table below shows the blood group phenotypes resulting from crosses between different genotypes.

Genotype of 1 st parent	Genotype of 2 nd parent	
	I ^A I ^A	I ^A I ^O
I ^A I ^O	A	A and O
I ^B I ^O	A and AB	A, B, O and AB

Two parents have a son who has blood group A and phenylketonuria. One parent has blood group O and the other has blood group AB. Neither parent has phenylketonuria.

What is the probability that the second child of these parents will be a girl with blood group B who does not have phenylketonuria?

- A 1 in 16
- B 1 in 8
- C 3 in 16
- D 3 in 8

23. In pea-plants, flower color and height are determined by two pairs of alleles. The genes for red color and tallness are dominant. Two flowers were crossed, and the offspring (F₁) were self-pollinated.

The results in the F₂ generation were as follows: 280 tall red; 90 tall white; 94 dwarf red; 30 dwarf white. Which one of the following is a possible description of the **parental generation**?

- A heterozygous tall red x heterozygous dwarf white
- B homozygous tall red x homozygous dwarf white
- C heterozygous tall red x homozygous dwarf white
- D homozygous tall white x heterozygous dwarf red



24. In a species of plant, the allele for hairy stems is dominant to the allele for smooth stems. The allele for purple stems is dominant to the allele for green stems.

A plant was self-fertilized and the offspring appeared in the ratio;

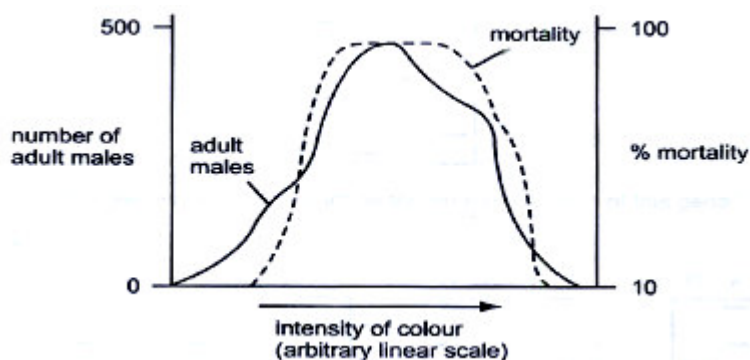
9 hairy, purple stem: 3 hairy, green stem: 3 smooth, purple stem: 1 smooth, green stem.

The offspring with smooth, green stems were crossed to all the other offspring.

What proportion of these crosses would be expected to produce offspring in the ratio 1 hairy, purple stem: 1 hairy, green stem: 1 smooth, purple stem: 1 smooth, green stem?

- A 9/15
- B 6/15
- C 4/15
- D 3/15

25. The graph below shows data on a population of a species of moth which shows considerable variation in color intensity.

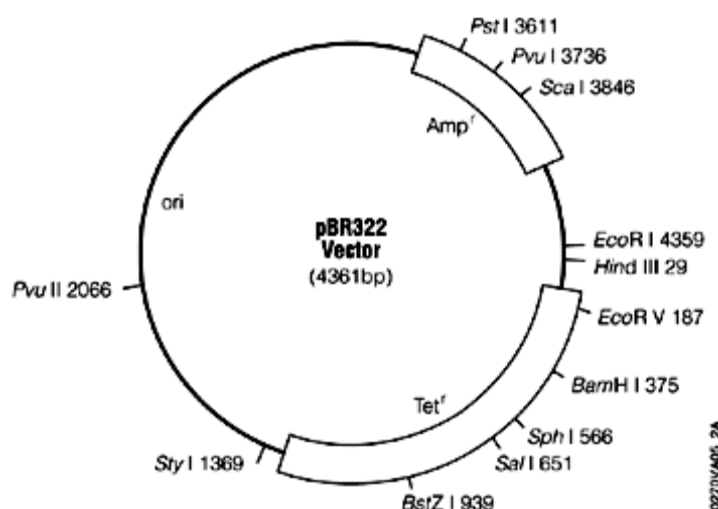


Which conclusion can be made from this graph?

- A Color variation is environmentally induced.
- B Color variation is genetically determined.
- C Extreme forms are favored by natural selection.
- D The species shows discontinuous variation with respect to color.



26. Which of the following would provide the best information for distinguishing phylogenetic relationships between several species that are almost identical in anatomy?
- A The fossil records.
 - B Homologous structures.
 - C Comparative embryology.
 - D Molecular comparisons of DNA and amino acid sequences.
27. pBR322 vector is used to clone a eukaryotic gene which has been digested by the restriction endonuclease *Bam*H1.



Following transformation, bacterial cells were grown in four different media as shown below;

I	Nutrient broth plus ampicillin
II	Nutrient broth plus tetracycline
III	Nutrient broth plus ampicillin & tetracycline
IV	Nutrient broth without antibiotics

Which of the following media would bacterial cells that contain the recombinant plasmids grow in?

- A I and II
- B I and III
- C I and IV
- D IV only



28. A student performed the following steps in a Southern blot experiment to determine the number of copies of a particular gene that has been inserted in a genetically modified organism.

- (i) Transfer of DNA to nitrocellulose membrane.
- (ii) Restriction digestion of genomic DNA.
- (iii) Cleaved DNA separated using gel electrophoresis.
- (iv) Create radioactive probe.
- (v) Incubate probe and membrane.

Which is the correct sequence to the above steps?

A	ii → iii → i → iv → v.
B	ii → iii → i → v → iv.
C	iv → v → i → ii → iii.
D	v → iv → iii → ii → i.

29. What are the normal functions of stem cells in a living human?

	Functions		
A	differentiating into many kinds of cells in a 3 -5 day old human embryo	producing insulin in pancreas damaged by type 1 diabetes	cells that can differentiate into bone cells in the skeleton
B	differentiating into many kinds of cells in a 3 -5 day old human embryo	producing red blood cells worn out by normal wear and tear	cells that can differentiate into cardiac muscles in the heart
C	differentiating into only one kind of cells in a 3 -5 day old human embryo	producing dopamine in brains of people with Parkinson's disease	cells that can differentiate into cartilage cells in the joint
D	differentiating into only one kind of cells in a 3 -5 day old human embryo	producing differentiated cells that can be used to screen new drugs	cells that can differentiate into nerve cells in the brain



30. Plants are more readily manipulated by genetic engineering than are animal cells because;

- A** plant genes do not contain introns.
- B** more vectors are available for transferring recombinant DNA into plant cells.
- C** a somatic plant cell can often give rise to a complete plant.
- D** plant cells have larger nuclei.

